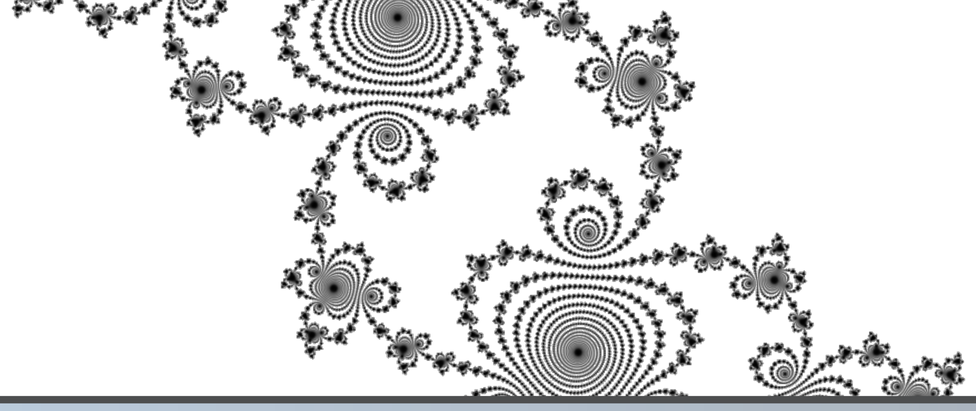

The Julia Set

Fractal Geometry

What is this?

Sometimes mathematicians touch on something so strange that the theory quickly stumbles out of the world of math, into the world of chaos and art. This is what happened with the Julia set.

The Theory

Predicable Madness

⇒ **Set** A set is a group of distinct objects. In this case, we are interested in terminating decimals, as they can be represented with binary code.

⇒ **Complex Numbers, C** A set where each element is represented as followed: $c \in \mathbb{C} \Rightarrow c = cos(x) + i * sin(y)$ for some $x, y \in \mathbb{R}$, and where $i^2 = -1$.

⇒ **Fractal** One often cited description that Mandelbrot published to describe geometric fractals is "a rough or fragmented geometric shape that can be split into parts, each of which is (at least approximately) a reduced-size copy of the whole".

⇒**Formal Definition of a julia set** If f is an entire function, then J(f) is the boundary of the set of points which converge to infinity under iteration.

Generating rules for our fractal of interest: the Quadratic Julia Set

1. Here we will concern ourselves with the complex quadratic polynomials.
2. $K(f_c) := \{z \in \mathbb{C} : \forall n \in \mathbb{N}, |f_c^n(z) < R\}$.
3. Here, $R > 0$ is a fixed real number.
4. $|R^2 - R| \geq |c|$, where c is some complex number.
5. $f_c(z) = z^2 + c$, where $z \in \mathbb{C}$.

⇒ **Some surprising facts** Fractals have the (abstract) property that on their limit points, there is a infinite length, yet their total surface area is in fact finite.

How do we program this?

1. We are doing calcuations on the complex plane, so we will need to map the reals to their complex counterparts. $x_c = x^2 - y^2$ and $y_c = 2xy$.
2. The distance between two values z_1, z_2 is given as $d_{12} = \sqrt{(x_{c1} - x_{c2})^2 + (y_{c1} - y_{c2})^2}$.
3. An element of our plane lives in the Julia set if, as $n- > \infty$, the function $f^n(x, y) < R$, for some R.
4. We want to loop through all such points x, y.

Code

Repository for the code of this project

https://github.com/stackedflows/C/blob/main/DISLIN/julia.c

Console Output

The dragon set: c = 0.8 + 0.156i